

Program: Diploma in Robotics Process Automation	
Course Code: 6309A	Course Title: Cyber Security Lab
Semester: 6	Credits: 1.5
Course Category: Programme Elective Course	
Periods per week: 3 (L: 0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Troubleshooting using software utility/commands.
- Troubleshooting using packet analyser tools.
- Setup and configure firewalls
- Setup an intrusion detection system.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Develop Python programs to solve simple problems	1008	Introduction to IT Systems Lab	I

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize with the working of Wireshark and study the different network troubleshooting commands.	15	Applying
CO2	Study of the features of firewall in providing network security and to set Firewall Security in windows.	6	Applying

CO3	Demonstrate the use of passwords in a computer system.	9	Applying
CO4	Demonstrate intrusion detection system using any tool (snort).	9	Applying
	LabTest	6	

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3	2					
CO4	3	2					

3-Stronglymapped, 2-Moderatelymapped, 1-Weaklymapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Familiarize with the working of Wireshark and study the different network troubleshooting commands.		
M1.01	Analysis Network using Wireshark for (a)Traffic Monitoring (b) Packet Sniffing	3	Applying
M1.02	Study the use of TCPDump and NetDiscover commands	6	Applying
M1.03	Familiarization of network trouble shooting commands - ping, traceroute/tracert, ipconfig/ifconfig, nslookup,netstat, putty/tera term,subnet and IP calculator, speedtest.net /pingtest.net ,pathping / mtr route , iptables, tcpdump, nmap, arp, dig	6	Applying
CO2	Study of the features of firewall in providing network security and to set Firewall Security in windows.		

M2.01	Study the steps to configure windows firewalls. Add exceptions.	2	Applying
M2.02	Firewall customization	1	Applying
M2.03	Access the Windows Firewall with Advanced Security. Configure inbound and outbound rules and analyse the traffic	3	Applying
	Lab Test–I	3	
CO3	Demonstrate the use of passwords in a computer system.		
M3.01	Study the steps to protect your personal computer system by creating User Accounts with Passwords and types of User Accounts for safety and security.	3	Applying
M3.02	Study the steps to protect a Microsoft Word Document using passwords, and removing passwords from a Microsoft Word Document	6	Applying
CO4	Demonstrate intrusion detection system using any tool (snort).		
M4.01	Installation of snort	3	Applying
M4.02	Demonstrate three modes of operation of snort	6	Applying
	Lab Test– II	3	

**** - Micro Project**

Students are expected to do a micro project in cyber security during the course for the purpose of continuous evaluation. This experiment shall be included in the bonafide record.

Example projects include:

- Implement various security attacks such as DoS attacks, SQL injection etc.
- Install an antivirus tool and set traffic rules.

Texts/References

T/R	BookTitle/Author
R1	Practical computer network analysis and design by McCabe, James D
R2	Computer Networking: A Top-Down Approach Kurose, Ross
R3	Wireshark Network Analysis (Second Edition): The Official Wireshark Certified Network Analyst Study Guide by Laura Chappell (Author), Gerald Combs (Foreword)

OnlineResources

Sl.No	WebsiteLink
1	http://vlabs.iitkgp.ac.in/ant
2	https://ocean.sagepub.com/online-network-analysis-voson-lab

List of Experiments:

1. Install wireshark and analyse network traffic for traffic monitoring.
2. Using wireshark, analyse network traffic for packet sniffing.
3. Study the use of TCPDump and NetDiscover commands
4. Demonstrate the network trouble shooting commands - ping, traceroute/tracert, ipconfig/ifconfig, nslookup, netstat, putty/teraterm, subnet and IP calculator, speedtest.net /pingtest.net, pathping / mtr route , iptables, tcpdump, nmap, arp, dig
5. Configure windows firewall and customize it.
6. Add exceptions in firewall and demonstrate the working.
7. Access the Windows Firewall with Advanced Security. Configure inbound and outbound rules and analyse the traffic
8. Protect a computer system by creating User Accounts with Passwords and types of User Accounts for safety and security.
9. Protect a Microsoft Word Document using passwords, and remove password from a Microsoft Word Document.
10. Install snort intrusion detection system and demonstrate its working
11. Configure three modes of operation in snort.